
ProteinGym: Large-Scale Benchmarks for Protein Fitness Prediction and Design

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Abstract

Predicting the effects of mutations in proteins is critical to many applications, from understanding genetic disease to designing novel proteins that can address our most pressing challenges in climate, agriculture and healthcare. Despite a surge in machine learning-based protein models to tackle these questions, an assessment of their respective benefits is challenging due to the use of distinct, often contrived, experimental datasets, and the variable performance of models across different protein families. Addressing these challenges requires scale. To that end we introduce ProteinGym, a large-scale and holistic set of benchmarks specifically designed for protein fitness prediction and design. It encompasses both a broad collection of over 250 standardized deep mutational scanning assays, spanning millions of mutated sequences, as well as curated clinical datasets providing high-quality expert annotations about mutation effects. We devise a robust evaluation framework that combines metrics for both fitness prediction and design, factors in known limitations of the underlying experimental methods, and covers both zero-shot and supervised settings. We report the performance of a diverse set of over 70 high-performing models from various subfields (eg., alignment-based, inverse folding) into a unified benchmark suite. We open source the corresponding codebase, datasets, MSAs, structures, model predictions and develop a user-friendly website that facilitates data access and analysis.

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1 Introduction

Proteins carry out a wide range of functions in nature, facilitating chemical reactions, transporting molecules, signaling between cells, and providing structural support to cells and organisms. This astonishing functional diversity is uniquely encoded in their amino acid sequence. For instance, the number of possible arrangements for a 64-residue peptide chain (20^{64}) is already larger than the estimated number of atoms in the universe. Despite substantial progress in sequencing over the past two decades, we have observed a relatively small, biased portion of that massive sequence space. Consequently, the ability to manipulate and optimize known sequences and structures represents tremendous opportunities to address pressing issues in climate, agriculture and healthcare.

The design of novel, functionally optimized proteins presents several challenges. It begins with learning a mapping between protein sequences or structures and their resulting properties. This mapping is often conceptualized as a “fitness landscape”, a multivariate function that characterizes the relationship between genetic variants and their adaptive fitness. The more accurately and comprehensively we can define these landscapes, the better our chances of predicting the effects of mutations and designing proteins with desirable and diverse properties. Machine learning, by modeling complex, high-dimensional relationships, has emerged as a powerful tool for learning these fitness landscapes. In recent years, a plethora of machine learning methods have been proposed for protein modeling, each promising to offer new insights into protein function and design. However, assessing the effectiveness of these methods has proven challenging. A key issue is their evaluation on distinct and relatively sparse benchmark datasets, while relative model performance fluctuates importantly across experimental assays, as was shown in several prior analyses [Riesselman et al., 2018, Laine et al., 2019, Meier et al., 2021]. This situation underscores the importance of scale in the benchmarks used. Larger, more diverse datasets would offer a more robust and comprehensive evaluation of model performance.

To address these limitations, we introduce ProteinGym, a large-scale set of benchmarks specifically tailored to protein design and fitness prediction. It comprises a broad collection of over 250 standardized Deep Mutational Scanning (DMS) assays which include over 2.7 million mutated sequences across more than 200 protein families, spanning different functions, taxa and depth of homologous sequences. It also encompasses clinical benchmarks providing high-quality annotations from domain experts about the effects of $\sim 65k$ substitution and indel mutations in human genes (§ 3).

We have designed ProteinGym to be an effective, holistic, robust, and user-friendly tool. It provides a structured evaluation framework that factors in known limitations of the underlying experimental methods and includes metrics that are tailored to protein design and mutation effect prediction (§ 4). We report the performance in a unified benchmark of over 70 diverse high-performing models that come from various subfields of computational biology (eg., mutation effects prediction, sequence-based models for de novo design, inverse folding), thereby supporting novel comparisons across. Unlike prior benchmarks, ProteinGym integrates both the zero-shot and supervised settings, leading to new insights (§ 5). All models are codified with a common interface in the same open-source codebase, promoting consistency and ease of use. Lastly, a dedicated website offers an interactive platform to facilitate comparisons across datasets and performance settings.

2 Related Work and Background

Multi-task protein benchmarks In recent years, several benchmarks have been introduced to provide initial means to assess protein model performance across a multitude of tasks of interests, e.g., predicting contacts, structure, thermostability, and fitness. These benchmarks are generally geared towards assessing the quality of learned protein representations, and the extent to which these representations can be broadly leveraged for various tasks. However, for fitness prediction, they all rely on a very limited set of proteins (e.g., 1-3 assays). In comparison, the ProteinGym benchmarks focus on a single task – fitness prediction – and encompass *two orders of magnitude more* point mutations assessed and vast diversity of protein families included.

TAPE (Tasks Assessing Protein Embeddings) [Rao et al., 2019] covers five protein prediction tasks, each designed to test a different aspect of protein function and structure prediction (secondary structure, contact, remote homology, fluorescence and stability), and focuses on assessments in the semi-supervised regime via carefully curated train-validation-test splits. ProteinGLUE [Capel et al.,

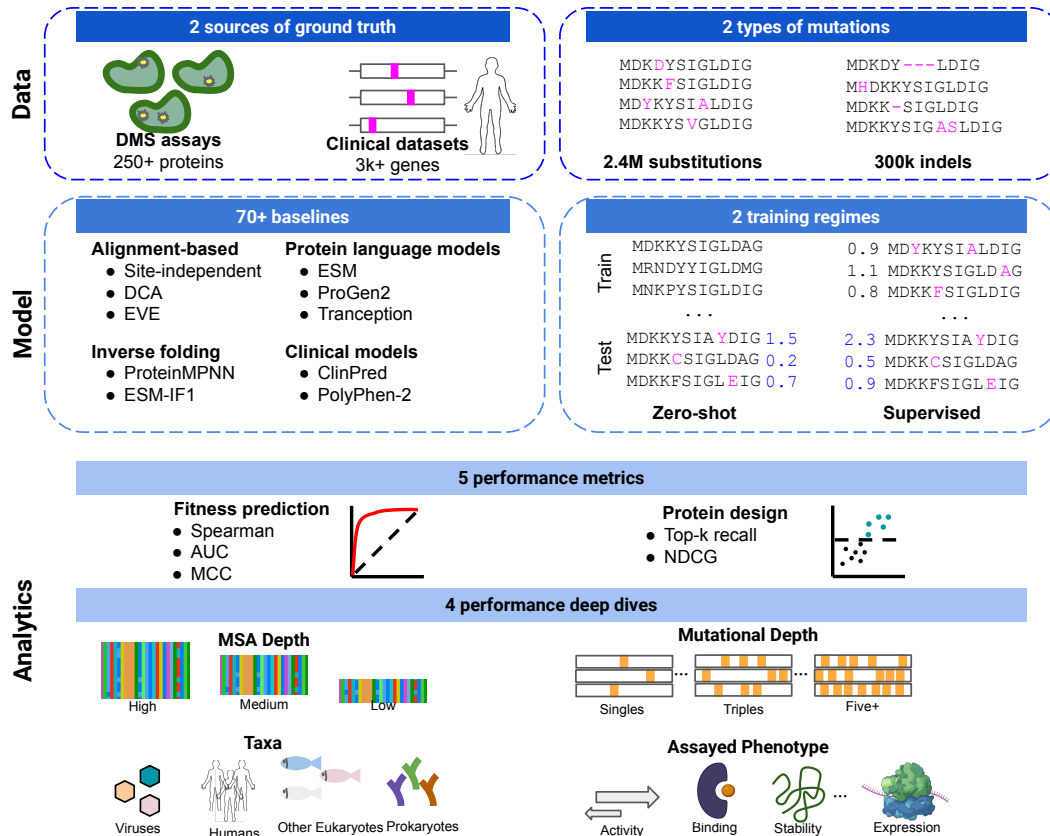


Figure 1: **ProteinGym benchmarks** ProteinGym is comprised of three layers. The data layer encompasses two complementary ground truth labels from DMS assays and clinical annotations from experts. For both, we analyze two types of mutations: substitutions and indels. The model layer is comprised of a diverse set of baselines, tailored to both zero-shot and supervised training regimes. Lastly, the analytics layer includes several performance metrics geared towards fitness prediction or protein design evaluation. Different segmentation variables (e.g., MSA depth, assayed phenotype, taxa) facilitate the comparisons of models across diverse settings

2022] also focuses on assessing the usefulness of learned protein representations on supervised downstream tasks. It is comprised of five different tasks, none directly related to protein fitness: secondary structure, solvent accessibility, protein-protein interactions, epitope region and hydrophobic patch prediction. PEER [Xu et al., 2022] also focuses on multi-task benchmarking, grouped in five categories: protein property, localization, structure, protein-protein interactions and protein-ligand interactions. It contains a richer set of evaluations compared with the prior two benchmarks, and also investigates the multi-task learning setting, but is not designed for thorough fitness prediction benchmarking (3 fitness related assays). The handful of fitness DMS assays from these various benchmarks are all subsumed in ProteinGym.

Single task, non-fitness datasets & benchmarks Efforts to create fair, large-scale, and comprehensive benchmarks have been a significant focus of computational biologists for certain tasks. Among these, the biennial Critical Assessment of protein Structure Prediction (CASP) [Kryshtafovych et al., 2021] is the most renowned. CASP concentrates on protein structure prediction and has set the gold standard in this domain. In parallel to CASP, the Critical Assessment of Functional Annotation (CAFA) [Zhou et al., 2019] challenge provides a platform for evaluating protein function classification. The SKEMPI [Moal and Fernández-Recio, 2012] database is specifically designed to aid the evaluation of computational methods predicting the effect of mutations on protein-protein binding affinity. Several datasets have been curated for specific properties of interest across a diverse set of

proteins, for instance thermostability [Tsuboyama et al., 2023, Stourac et al., 2020, Chen et al., 2022] or solubility [Hon et al., 2020].

Protein fitness benchmarks Closest to our work are the collections of DMS assays that were curated in Hopf et al. [2017] (28 substitution assays), and then further expanded upon in Riesselman et al. [2018] (42 substitution assays) and Shin et al. [2021] (4 indel assays). We include all assays related to fitness prediction from these prior works in ProteinGym. FLIP [Dallago et al., 2021] focused on comparing fitness predictors in the semi-supervised setting, developing a robust evaluation framework and curating cross-validation schemes for three assays. MaveDB [Rubin et al., 2021] is a repository rather than a benchmark, but it compiles a large collection of datasets from multiple variant effect mapping experiments that can be used for benchmarking purposes. An initial prototype of the ProteinGym benchmarks (referred to as ‘ProteinGym v0.1’) was introduced in Notin et al. [2022a]. We have since then significantly expanded the benchmarks in terms of number and diversity of underlying datasets, baselines, evaluation framework and model training regimes (Table A1). This not only enables performance evaluation at an unprecedented scale, but also builds connections between different subfields that are often perceived as separate, as we discuss in the following paragraph.

Clinical Benchmarks Designing an unbiased, non-circular and broadly applicable benchmark to evaluate the performance of human variant effect predictors at predicting clinical significance is still an open-problem for the clinical community. Combining DMS with clinical annotations has been a fruitful direction to avoid biases [Frazer et al., 2021, Livesey and Marsh, 2023]. ClinGen curated a clinical dataset specifically designed to compare a subset of models [Pejaver et al., 2022].

Relationship between protein fitness, mutation effect prediction and design The protein fitness landscape refers to the mapping between genotype (e.g., the amino acid sequence) and phenotype (e.g., protein function). While it is a fairly broad concept, it should always be thought about in practice within a particular context (e.g., stability at a given temperature in a specific organism). Models that learn the protein fitness landscape have been shown to be effective at predicting the effects of mutations [Frazer et al., 2021, Jagota et al., 2022, Brandes et al., 2023, Notin et al., 2022b]. But the ability to tell apart the sequences that are functional or not is also critical to protein engineering efforts [Romero et al., 2012, Yang et al., 2018, Wu et al., 2019, Alley et al., 2019b]. Although typically introduced in the context of de novo protein design [Huang et al., 2016], inverse folding methods [Ingraham et al., 2019, Jing et al., 2020, Dauparas et al., 2022, Gao et al., 2022] can also be used for mutation effects prediction (Appendix A.4.1). There is thus a very tight connection between protein fitness, mutation effect prediction and protein engineering, and the same models can be used for either task depending on context. We seek to illustrate this connection through this work, comparing baselines introduced in different fields (e.g., protein representation learning, inverse folding models, co-evolution models) on the same benchmarks, and including different metrics that are geared more to mutation effect prediction (e.g., Spearman) or design tasks (e.g., NDCG).

3 ProteinGym benchmarks

ProteinGym is a collection of benchmarks (Fig. 1) that cover different types of mutation (ie., substitutions vs. indels), ground-truth labels (ie., experimental measurement from DMS vs. clinical annotations), and model training regime (ie., zero-shot vs. supervised).

3.1 Mutation types

We curate benchmarks for two types of protein mutations – *substitutions* and *indels* (insertions or deletions), each with unique implications for the structure, function, and modeling of proteins.

Substitutions Substitution mutations refer to a change in which one amino acid in a protein sequence is replaced by another. Depending on the properties of the substituted amino acid, this can have varied impacts on the protein’s structure and function, which can range from minimal to drastic. The influence of a substitution largely depends on whether it is conservative (i.e., the new amino acid shares similar properties to the original) or non-conservative. In terms of computational modeling, substitutions are the most commonly addressed mutation type, and the majority of mutation effect predictors support substitutions.

Indels Indel mutations correspond to insertions or deletions of amino acids in protein sequences. While indels can affect protein fitness in similar ways to substitutions, they can also have profound impacts on protein structure by altering the protein backbone, causing structural modifications inaccessible through substitutions alone [Shortle and Sondek, 1995, Tóth-Petróczy and Tawfik, 2013]. From a computational perspective, indels present a unique challenge because they alter the length of the protein sequence, requiring additional considerations in model design and making it more difficult to align sequences. For instance, the majority of models trained on Multiple Sequence Alignments are typically unable to score indels due to the fixed coordinate system they operate within (see § 4). Furthermore, when dealing with probabilistic models, comparing relative likelihoods of sequences with different lengths results in additional complexities and considerations.

3.2 Dataset types

The fitness of a protein is a measure of how well a protein can perform its function within an organism. Factors that influence protein fitness are diverse and include stability, folding efficiency, catalytic activity (for enzymes), binding specificity and affinity. To properly capture this diversity, we curated a broad set of experimental assays that map a given sequence to phenotypic measurements that are known or hypothesized to be related to its fitness. We focused on two potential sources of ground truth: Deep Mutational Scanning (DMS) assays and Clinical datasets.

Deep Mutational Scanning assays Modeling protein fitness landscapes presents a challenge due to the complex relationship between experimentally measured protein fitness, the distribution of natural sequences, and the underlying fitness landscape. It is challenging to isolate a singular, measurable molecular property that reflects the key aspects of fitness for a given protein. In developing ProteinGym, we prioritized assays where the experimentally measured property for each mutant protein is likely to represent the role of the protein in organismal fitness. The resulting compilation of over 250 DMS assays extends over a wide array of functional properties, including ligand binding, aggregation, thermostability, viral replication, and drug resistance. It encompasses diverse protein families, such as kinases, ion channel proteins, G-protein coupled receptors, polymerases, transcription factors, and tumor suppressors. In contrast to most DMS assay collections that focus exclusively on single amino acid substitutions, ProteinGym includes several assays with multiple amino acid variants. Moreover, it spans different taxa (i.e., humans, other eukaryotes, prokaryotes, and viruses), alignment depths, and mutation types (substitutions vs indels). All details about the curation and pre-processing of these DMS assays are provided in Appendix A.3.

Clinical datasets ClinVar [Landrum and Kattman, 2018] is an extensive, public database developed by the National Center for Biotechnology Information (NCBI). It serves as an archival repository that collects and annotates reports detailing the relationships among human genetic variations and associated phenotypes with relevant supporting evidence, thereby providing robust, clinically annotated datasets that are invaluable for understanding the functional impact of mutations. From the standpoint of benchmarking mutation effects predictors, ClinVar permits the direct comparison of predictive models in terms of their accuracy in estimating the functional impact of mutations on human health. Annotations are also available for an order of magnitude more distinct proteins compared with our DMS-based benchmarks, albeit much sparser per protein (see table 1). In the case of indels, we focused on short (≤ 3 amino acids) variants. In ClinVar, 84% of indel annotations are pathogenic, so we added to our clinical dataset common indels from gnomAD (allele frequency $>5\%$) as pseudocontrols [Karczewski et al., 2020].

3.3 Model training regime

Lastly, we discriminate in our benchmarks between zero-shot and supervised settings. In the supervised regime we are allowed to leverage a subset of labels to train a predictive model, while in the zero-shot setting we seek to predict the effects of mutations on fitness without relying on the ground-truth labels for the protein of interest. These two settings offer complementary viewpoints of practical importance. For instance, in settings where labels are subject to several biases or scarcely available (e.g., labels for rare genetic pathologies), we need methods with robust zero-shot performance. In cases where we seek to design new proteins that simultaneously optimize several properties of interest (e.g., binding affinity, thermostability) and we have collected a sufficiently large number of labels for each target, supervised methods are more appropriate. The